

Pattern Recognition and Machine Learning (EE602L)

Lab 1: Data Exploration and Preparation

Instructions:

Use Google Colab.

Submit one well-commented notebook (Lab1-RollNum.ipynb) along with a short PDF report (Lab1-RollNum.pdf) containing your outputs and explanations. Wrong file names or missing files won't be graded.

Deadline: Monday, 17 August, 11:59 PM. Late submissions lose 10% per day.

This is an individual assignment. Please follow the Institute's Academic Code of Honor.

Dataset:

insurance.csv

Download: <https://raw.githubusercontent.com/stedy/Machine-Learning-with-R-datasets/master/insurance.csv>

Questions:

Using the insurance.csv dataset, answer the following questions:

Q1. Load the dataset and print its shape, column names, data types and summary statistics (.describe()). Identify which columns form the feature vector x and which is the label y . State N and d (count d as the number of raw feature columns, before any encoding), and whether this is a regression or classification problem, and why. Write one line connecting this dataset to the $D = \{(x_i, y_i)\}$ formulation.

Q2. Compute the average charges grouped by smoker, sex and region — which factor seems to matter most? Base this on the magnitude of the difference in mean charges across groups; no formal hypothesis test is required. Compute the correlation of age, bmi and children with charges and interpret the numbers briefly. Then plot: a histogram of charges, a boxplot of charges by smoker, a scatter plot of bmi vs charges coloured by smoker, a correlation heatmap of the numeric columns, and a bar chart of average charges by region. In 3–4 sentences, summarise what the plots and numbers tell you about the data.

Q3. Engineer at least 4 new features from the raw data - for example a BMI category, an age group, a smoker $\times$ BMI interaction flag, family size, a BMI $\times$ age interaction, or a senior indicator. For each, write 1–2 lines on why it might help predict charges better than the raw columns. Show the first 10 rows of your updated dataframe. Do not create features using charges or any quantity derived from charges (e.g. average charges per region/group) — this constitutes target leakage. This updated dataframe (raw columns plus your engineered features) is what you will split in Q4 and continue using through Q6.

Q4. Write a function custom_split(df, train_ratio=0.7, val_ratio=0.15, test_ratio=0.15, seed=42) that shuffles and splits the dataset using only NumPy and Pandas — no sklearn.model_selection. Print the size of each split and verify programmatically that there's no overlap (no data leakage) between the three sets.

Q5. The dataset is fairly clean, so first corrupt your own copy of it (create it from your Q3 dataframe, without modifying that dataframe itself): introduce at least 5 missing values across bmi/age, at least 3 duplicate rows, inconsistent text formatting in sex or region (e.g. 'male', 'Male ', 'MALE'), and at least

one extreme outlier in charges or bmi. Use your own seed so your version is unique, and print/state that seed for reproducibility. Then write a script that handles the missing values (justify mean/median/mode or dropping), removes duplicates, standardizes the text columns, detects the outlier(s) using the IQR rule or z-score (state which method you chose) and handles them, and prints a short before/after summary. This is an independent data-cleaning exercise — it does not feed into Q6. For Q6, continue using your uncorrupted Q3 dataframe (not this corrupted copy) and the splits from Q4.

Q6. Using the splits from Q4, scale the input features age and bmi — either min-max or z-score, your choice (you may use `sklearn.preprocessing` for this; the Q4 restriction applies only to `sklearn.model_selection`) — but compute the scaling parameters from the training set only, and apply them to validation and test. Do not scale charges, since it is the target y , not an input feature. In 3–4 sentences, explain why the scaler shouldn't be fit on the full dataset before splitting, and what could go wrong if it were. Plot the distribution of training-set bmi before and after scaling.